Frequency Selective Circuits and Bode Diagrams

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1. Bode Diagrams

1.1. Real, First-Order Poles and Zeros

We have the expression of $H(j\omega)$ in *standard form* and let K_0 represent the constant quantity Kz_1/p_1 and at the same time we express $H(j\omega)$ in polar form

$$H(j\omega) = \frac{Kz_1(1 + j\omega/z_1)}{p_1(j\omega)(1 + j\omega/p_1)} \to H(j\omega) = \frac{K_0\,|1 + j\omega/z_1|}{|\omega||1 + j\omega/p_1|}\underline{/\psi_1 - 90^\circ - \beta_1}$$

then we got

$$|H(j\omega)| = \frac{K_0|1 + j\omega/z_1|}{\omega|1 + j\omega/p_1|}, \qquad \theta(\omega) = \psi_1 - 90^\circ - \beta_1$$

By definition, the phase angles are

$$\psi_1 = \tan^{-1}\omega/z_1, \quad \beta_1 = \tan^{-1}\omega/p_1$$

The Bode diagrams consist of plotting $|H(j\omega|)$ (amplitude) and $\theta(\omega)$ (phase) as functions of ω

1.2. Straight-Line Amplitude Plots

The amplitude of $H(j\omega)$ in decibels is

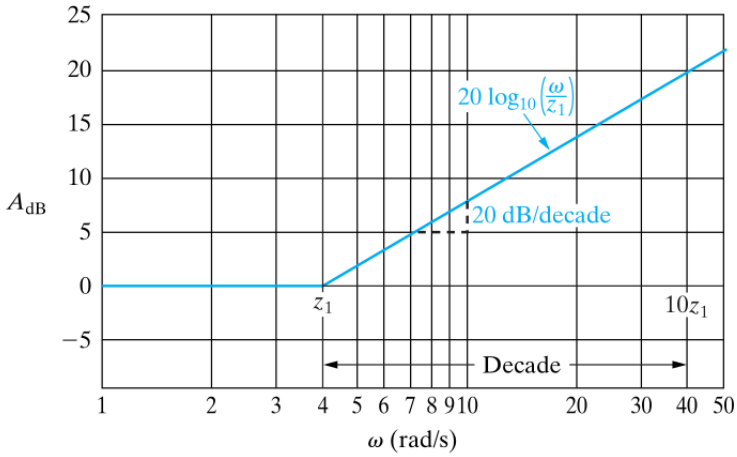
$$A_{dB} = 20\log_{10}|H(j\omega)|$$

$$A_{dB} = 20\log_{10}K_0 + 20\log_{10}|1 + j\omega/z_1| - 20\log_{10}\omega - 20\log_{10}|1 + j\omega/p_1|$$

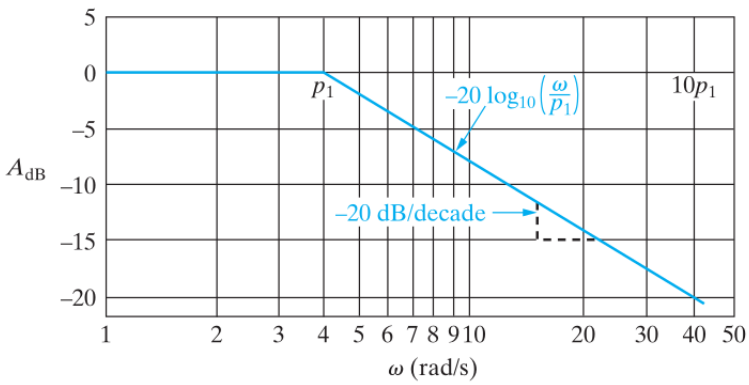
The plot of $20\log_{10}K_0$ is a horizontal straight line because K_0 is not a function of frequency. The value of this term is positive for $K_0 > 1$, zero for $K_0 = 1$, and negative for $K_0 < 1$

$$20\log_{10}|1 + j\omega/z_1| \to 0 \text{ as } \omega \to 0$$

$$20\log_{10}|1 + j\omega/z_1| \to 20\log_{10}(\omega/z_1) \text{ as } \omega \to \infty$$



On a log frequency scale, $20 \log_{10}(\omega/z_1)$ is a straight line with a slope of 20 dB/decade (a decade is 10-to-1 change in frequency). This straight line intersects the 0 dB axis at $\omega = z_1$. This value of ω is called the *corner frequency*



The plot of $-20 \log_{10} \omega$ is a straight line having a slope of -20dB/decade that intersects the 0 dB axis at $\omega = 1$. Two straight lines approximate the plot of $-20 \log_{10} |1 + j\omega/p_1|$. Two straight line intersect at $\omega = p_1$. For larger values of ω, the straight line has a slope of -20 dB/decade.

1.3. More Accurate Amplitude Plots

We can make the straight-line plots for first-order poles and zeros more accurate by correcting the amplitude values at the corner frequency, one half the corner frequency, and twice the corner frequency.

At the corner frequency, the actual value in decibels is

$$A_{dB_c} = \pm 20 \log_{10} |1 + j1| \approx \pm 3 \text{ dB}$$

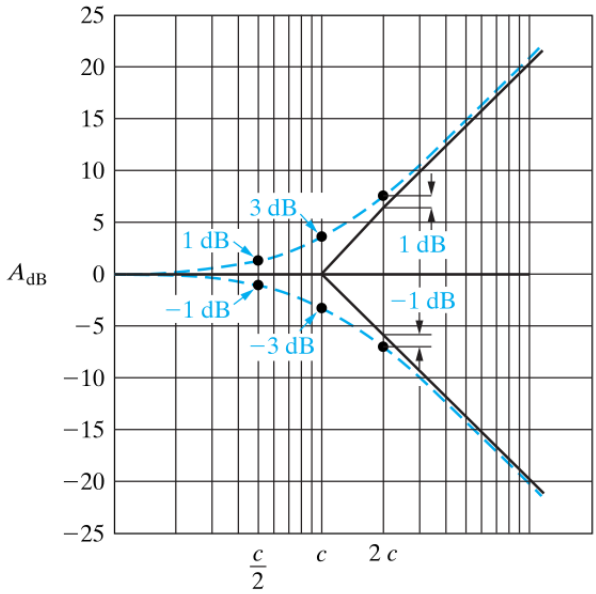
At one half the corner frequency is

$$A_{dB_{c/2}} = \pm 20 \log_{10} |1 + j\frac{1}{2}| \approx \pm 1 \text{ dB}$$

At twice the corner frequency, the actual value in decibels is

$$A_{dB_{2c}} = \pm 20 \log_{10} |1 + j2| \approx \pm 7 \text{ dB}$$

The plus sign applies to a first-order zero, and the minus sign applies to a first-order pole.

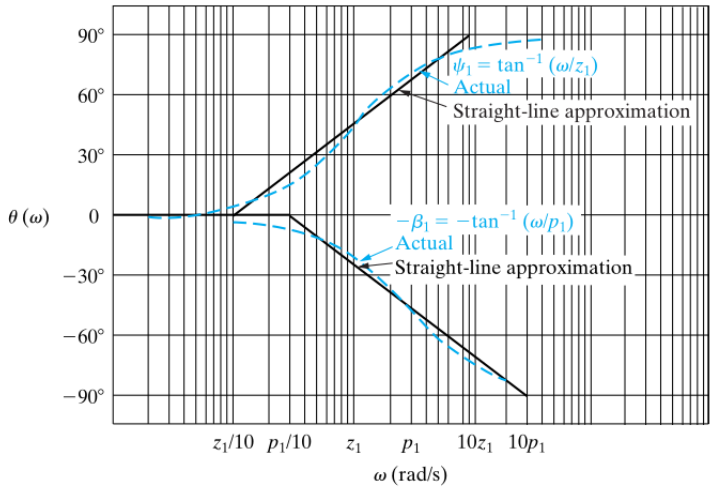


1.4. Straight-Line Phase Angle Plots

The phase angle associated with the constant K_0 is zero, and the phase angle associated with a first-order zero or pole at the origin is a constant $\pm 90°$. For a first-order zero or pole not at the origin, the straight-line approximations are as follows:

- For frequencies less than one tenth the corner frequency, the phase angle is assumed to be zero.
- For frequencies greater than 10 times the corner frequency, the phase angle is assumed to be $\pm 90°$
- Between one tenth the corner frequency and 10 times the corner frequency, the phase angle plot is a straight line that goes through $0°$ at one-tenth the corner frequency, $\pm 45°$ at the corner frequency, and $\pm 90°$ at 10 times the corner frequency.

In all these cases, the plus sign applies to the first-order zero and the minus sign to the first-order pole.



The maximum deviation between the straight-line plot and the actual plot is approximately 6°

1.5. Bode Diagrams: Complex Poles and Zeros

$$H(s) = \frac{K}{(s+\alpha-j\beta)(s+\alpha+j\beta)}$$

Rewrite the product $(s+\alpha-j\beta)(s+\alpha+j\beta)$ as $s^2+2\alpha s+\alpha^2+\beta^2$ (*)

When making Bode diagrams, we write the quadratic term in more convenient form $s^2+2\zeta\omega_n s+\omega_n^2$, a direct comparison of the two forms shows

$$\omega_n^2 = \alpha^2+\beta^2, \quad \zeta\omega_n = \alpha$$

If $\zeta < 1$ the roots of the quadratic factor are complex, and we use (*) to represent the complex poles. If $\zeta \geq 1$, we factor the quadratic factor into $(s+p_1)(s+p_2)$ and then plot amplitude and phase. Assuming that $\zeta < 1$ we got

$$H(s) = \frac{K}{s^2+2\zeta\omega_n s+\omega_n^2} \rightarrow H(j\omega) = \frac{K_0}{1-(\omega^2/\omega_n^2)+j(2\zeta\omega/\omega_n)}$$

with

$$K_0 = \frac{K}{\omega_n^2}$$

Replace the ratio ω/ω_n by u then write $H(j\omega)$ in polar form we got

$$A_{dB} = 20\log_{10}K_0 - 20\log_{10}|(1-u^2)+j2\zeta u|$$

and

$$\theta(\omega) = -\beta_1 = -tan^{-1}\frac{2\zeta u}{1-u^2}$$

1.6. Amplitude Plots

As $u \rightarrow 0, \omega \rightarrow 0$

$$A_{dB} \rightarrow 0$$

As $u \to \infty, \omega \to \infty$

$$A_{dB} \to -40 \log_{10} u$$

For $\omega < \omega_n$ the straight line lies along the 0 dB axis, and for $\omega > \omega_n$ the straight line has a slope of -40 dB/decade. These two straight lines join on the 0 dB axis at $u = 1$ or $\omega = \omega_n$

1.7. Correcting Straight-Line Amplitude Plots

For sketching pur- poses, the straight-line amplitude plot can be corrected by locating four points on the actual curve. These four points correspond to (1) one half the corner frequency, (2) the frequency at which the ampli- tude reaches its peak value, (3) the corner frequency, and (4) the fre- quency at which the amplitude is zero.

1. At one half the corner frequency

$$A_{dB}(\omega_n/2) = -10 \log_{10}(\zeta^2 + 0.5625)$$

2. The amplitude peaks at a frequency of $\omega_p = \omega_n \sqrt{1 - 2\zeta^2}$

$$A_{dB}(\omega_n) = -10 \log_{10} \left[4\zeta^2(1 - \zeta^2)\right]$$

3. At the corner frequency the actual amplitude is

$$A_{dB}(\omega_n) = -20 \log_{10} 2\zeta$$

4. the corrected amplitude plot crosses the 0 dB axis at

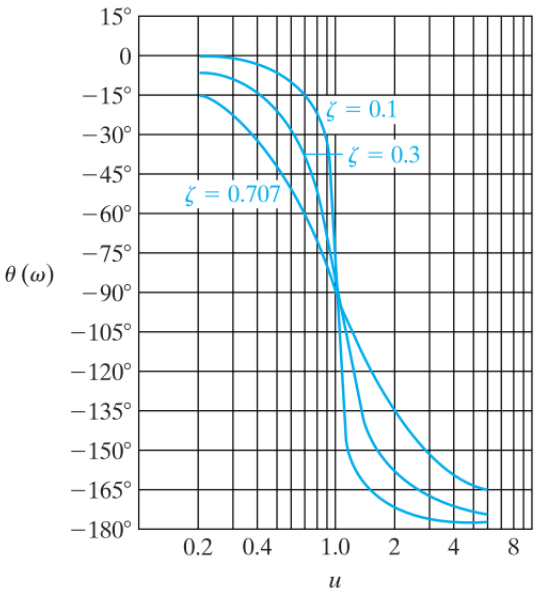
$$\omega_0 = \sqrt{2}\omega_p$$

1.8. Phase Angle Plots

The phase angle plot for a pair of complex poles is a plot of

$$\theta(\omega) = -\beta_1 = -tan^{-1}\frac{2\zeta u}{1-u^2}$$

The phase angle is zero at zero frequency and is $-90°$ at the corner frequency. It approaches $-90°$ as $\omega(u)$ becomes large. As in the case of the amplitude plot, ζ is important in determining the exact shape of the phase angle plot. For small values of ζ, the phase angle changes rapidly in the vicinity of the corner frequency. Figure below shows the effect of ζ on the phase angle plot.



2. Introduction to Frequency Selective Circuit

We begin this chapter by analyzing circuits from each of the four major categories of filters: low pass, high pass, band pass, and band reject. The transfer function of a circuit is the starting point for the frequency response analysis.

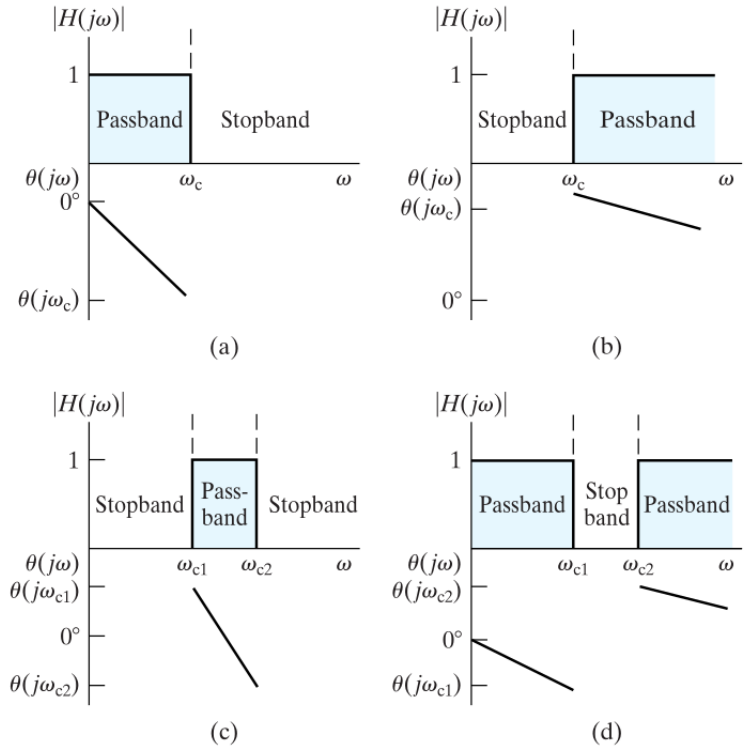
2.1. Preliminaries

The signals passsed from the input to the output fall within a band of frequencies called the *passband*. Input voltages outside this band have thier magnitudes attenuated by the circuit and are thus effectively prevented from reaching the output terminals of the circuit. Frequencies not in a circuit's passband are in its *stopband*. Frequency-selective circuits are categorized by the location of the passband.

One way of identifying the type of frequency-selective circuit is to examine a frequency response plot.

The ideal frequency response plots for the four major categories of filters. Parts (a) and (b) illustrate the ideal plots for a low-pass and a high-pass filter, respectively. Both filters have one passband and one stopband, which are defined by the cutoff frequency that separates them. The names low pass and high pass are derived from

the magnitude plots: a low-pass filter passes signals at frequencies lower than the cutoff frequency from the input to the output, and a high-pass filter passes signals at frequencies higher than the cutoff frequency.



(a) An ideal low-pass filter. (b) An ideal high-pass filter.
(c) An ideal bandpass filter. (d) An ideal bandreject filter.

The two remaining categories of filters each have two cutoff frequencies. (c) illustrates the ideal frequency response plot of a bandpass filter, which passes a source voltage to the output only when the source frequency is within the band defined by the two cutoff frequencies. (d) shows the ideal plot of a bandreject filter, which passes a source voltage to the output only when the source frequency is outside the band defined by the two cutoff frequencies. The bandreject filter thus rejects, or stops, the source voltage from reaching the output when its frequency is within the band defined by the cutoff frequencies.

2.2. Low-Pass Filters

Defining the Cutoff Frequency

The definition for cutoff frequency widely used by electrical engineers is the frequency for which the transfer function magnitude is decreased by the factor $1/\sqrt{2}$ from its maximum value

$$|H(j\omega_c)| = \frac{1}{\sqrt{2}} H_{max}$$

At the cutoff frequency ω_c, the average power delivered by the circuit is one half the maximum average power, Thus, ω_c is also called the *half-power frequency.* Therefore, in the passband, the average power delivered to a load is at least 50% of the maximum

average power

Cutoff frequency for RL filters

$$\omega_c = \frac{R}{L}$$

Cutoff frequency for RC filters

$$\omega_c = \frac{1}{RC}$$

Transfer function for a low-pass filter

$$H(s) = \frac{\omega_c}{s + \omega_c}$$

Examples

In a series RC circuit (where the output voltage is taken across the capacitor C) or a series RL circuit (where the output voltage is taken across the resistor R), the behavior depends on the frequency. At low frequencies, the capacitor C acts as an open circuit (or the inductor L acts as a short circuit), resulting in an output voltage that is approximately equal to the input voltage. Conversely, at high frequencies, the capacitor C becomes a short circuit (or the inductor L becomes an open circuit), causing the output voltage to attenuate toward zero.

2.3. High-pass Filters

Transfer function for a high-pass filter

$$H(s) = \frac{s}{s + \omega_c}$$

By simply swapping the component positions of a low-pass filter, a high-pass filter is formed. Specifically, a series RC circuit (with the output taken across R) or a series RL circuit (with the output across L) functions as a high-pass filter.

2.4. Bandpass Filters

Center Frequency, Bandwidth, and Quality Factor

There are three other important parameters that characterize a bandpass filter. he first is the center frequency, ω_o, efined as the frequency for which a circuit's transfer function is purely real. Another name for the center frequency is the resonant frequency. This is the same name given to the frequency that characterizes the natural response of the second-order circuits, because they are the same frequencies! When a circuit is driven at the resonant frequency, we say that the circuit is in resonance, because the frequency of the forcing function is the same as the natural frequency of the circuit. The center frequency is the geometric center of the passband, that is,

$$\omega_o = \sqrt{\omega_{c_1}\omega_{c_2}}$$

The second parameter is the bandwidth, β , which is the width of the pass- band.The final parameter is the quality factor, which is the ratio of the center frequency to

the bandwidth.The quality factor gives a measure of the width of the passband, independent of its location on the frequency axis. It also describes the shape of the magnitude plot, independent of frequency.

Series RLC Circuit

Center frequency

$$\omega_o = \sqrt{\frac{1}{LC}}$$

Cutoff frequencies, series RLC filters

$$\omega_{c_1} = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \left(\frac{1}{LC}\right)}$$

$$\omega_{c_2} = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \left(\frac{1}{LC}\right)}$$

Relationship between bandwidth and cutoff frequencies

$$\beta = \omega_{c_2} - \omega_{c_1}$$

Quality factor

$$Q = \omega_o/\beta$$

Transfer function for RLC bandpass filter

$$H(s) = \frac{\beta s}{s^2 + \beta s + \omega_o^2}$$

2.5. Bandreject Filters

This filter passes source voltages outside the band between the two cutoff frequencies to the output (the passband), and attenuates source voltages before they reach the output at frequencies between the two cutoff frequencies (the stopband). Bandpass filters and bandreject filters thus perform complementary functions in the frequency domain.

Transfer function for RLC bandreject filter

$$H(s) = \frac{s^2 + \omega_o^2}{s^2 + \beta s + \omega_o^2}$$

the numerator is reponsible for creating "Zero" points on the complex axis $\pm j\omega_o$, force signals to 0 at this specific frequency.